Essential in Inventor (40 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**GETTING STARTED** **Introduction:** Understanding parametric design, Understanding feature based concept, Basic modeling concepts, Understanding History-Based Modeling and Dependencies, Adaptivity, Orthographic Projections, File Extensions, Inventor Workflow GUI Mouse Control Functional Keys **View Control:** View Cube, Navigation Wheel, Pan, Zoom, Rotate, Look At, Previous View, Next View, Projections, Visual Styles, Clean Screen, Switch, Tile Window, Cascade New **Work Environment:** Project Files, Application Options, Document Settings, File Open Options	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for practice	Practicals	2.0 Hrs
3	Session 3	**2D SKETCHES** Sketch Workflow **Creating 2D Entities:** Line, Circle, Ellipse, Arc, Rectangle, Polygon, Fillet, Chamfer **Advanced Sketching:** Spline, Equation Curve, Slot, Sketch Point, Centre Point, Text & Geometry Text, Creating 3D Entities, Sketch Dimensioning, Sketch Constraints	Theory/Demo	2.0 Hrs
4	Session 4	Exercises included for practice	Practicals	2.0 Hrs
5	Session 5	**SKETCH BLOCK** **Modifying Sketch**: Move, Copy, Rotate, Trim, Extend, Split, Scale, Stretch, Offset, Editing Spline Sketch Pattern: Rectangular, Circular, Mirror Projection Sketch Properties	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for practice	Practicals	2.0 Hrs
7	Session 7	**REFERENCE PLANE** **Work Feature:** Work Plane, Work Axis, Work Point, UCS, in line work features **Part Modeling:** Overview on Part Modeling, Task Performed in Part Modeling, Part Browser, Part Template Primitives	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for practice	Practicals	2.0 Hrs
9	Session 9	**FEATURES** **Placed Features:** Hole, Thread, Fillet, Chamfer, Shell, Draft	Theory/Demo	2.0 Hrs
10	Session 10	Exercises included for practice	Practicals	2.0 Hrs
11	Session 11	**Advanced Created Features:** Derive, Emboss, Rib, Decal **Solid Editing:** Direct Edit, Combine, Split, Thicken/Offset **Pattern:** Rectangular, Circular, Mirror	Theory/Demo	2.0 Hrs
12	Session 12	Exercises included for practice	Practicals	2.0 Hrs
13	Session 13	**PART & ASSEMBLIES** **Measure:** Distance, Angle, Loop, Area, Region Properties Parameters Assembly Modeling Environment, Assembly Browser **Assembly Components:** Placements and Constraints, Assembly Relationship Error & Recovery Tools Flexible Components, Adaptive Components	Theory/Demo	2.0 Hrs
14	Session 14	Exercises included for practice	Practicals	2.0 Hrs
15	Session 15	**Working with Assembly Components:** Copy Assembly Components, Assembly Components in Patterns, Move & Rotation of Assembly Components, Deleting Assembly Components Replace Components Inspection for Interference	Theory/Demo	2.0 Hrs
16	Session 16	Exercises included for practice	Practicals	2.0 Hrs

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17	Session 17	**SHEET METAL DESIGN** Sheet Metal Defaults **Sheet Metal Features:** Face, Flange, Contour Flange, Lofted Flange, Contour Roll, Hem, Bend, Fold, Cut, Corner Seam, Rip, Unfold, Refold, Hole, Corner Round, Corner Chamfer, Copy Object, Punch Tool	Theory/Demo	2.0 Hrs
18	Session 18	Exercises included for practice	Practicals	2.0 Hrs
19	Session 19	**DETAILING** **Creating Fundamental Views:** Base View, Projected View, Auxiliary View, Section View, Detail View, Overlay View, Draft View Alignment, Creating Sketches in Drawing **Modifying Sketches:** Break View, Breakout View, Slice, Crop Creating New Sheet **Annotate:** Dimensioning, Arrange Dimension, Retrieve Dimension, Notes, Text, GD&T Symbols Tables, Balloon	Theory/Demo	2.0 Hrs
20	Session 20	Exercises included for practice	Practicals	2.0 Hrs